RESILIA MZ

Questions and answers in simple English

Short sentences and easy words, ready to say to the jury. Each answer has a **short version** in red: if you are nervous, say only that, then one number.

98.6 % of days with light all day	**17 h → 0 h** without light on a heavy-rain day	**81k vs 180k** USD: right mix vs batteries only	**100 %** solar energy

1. The two most important questions

1. What is the innovation of your project?

Solar panels and batteries are not new. Our innovation is **how we choose the size** of the system.

Most systems are made for a **normal day**. But on rainy days, the village has no light.

We test the system on **many real days** in Mabote, one after another: sunny, cloudy and rainy days.

Our rule is simple: **first, the light must not go off. Then, we choose the cheapest system.**

The result: the village has light on **98.6 % of days**, with only solar energy.

Short version: “We don't plan for the normal day. We plan for the bad days.”

2. Is the resilient system just more batteries?

No. A battery does not make energy. It only **stores** energy. The panels **make** the energy.

So we added **more panels** (30 to 55 kWp) and **a little more battery** (100 to 140 kWh).

We tested 156 systems. A small system has power cuts on **107 days a year**. Batteries only would cost **180 thousand dollars**. Our system costs **81 thousand dollars** and gives light on **98.6 % of days**.

It is not too big: on the heavy-rain day, the battery ends at **26 %**. The safe minimum is 20 %.

Short version: “Not more batteries. The right mix.”

2. Questions about the challenge

3. What does your tool do?

It shows, **hour by hour**, how a solar mini-grid gives energy to a village during 24 hours.

It shows when the sun gives energy, when the battery gives energy, and when the village has **no light**.

Then it finds the **right size** of panels and battery for the village.

Short version: "It shows the energy hour by hour and finds the right system size."

4. How does the system balance supply and demand in 24 hours?

Every hour, the same simple rule:

1) The sun gives energy to the village. 2) Extra energy goes into the battery. 3) At night, or when there is not enough sun, the battery gives the energy. 4) If the battery is empty, the village has no light.

Short version: "Sun first, then battery. If both are empty, no light."

5. What are the inputs and the outputs?

Inputs: the size of the solar panels, the size of the battery, the number of families and the services (health centre, school, water pump, grain mill). We also use the real sun and weather of Mabote.

Outputs: the energy balance hour by hour, the battery level, the hours without light, and the risk of power cuts.

Short version: "Inputs: panels, battery, families. Outputs: energy balance and risk of cuts."

6. The challenge talks about wind energy. Why do you use only solar?

We checked the real wind in Mabote. The average is only **3.3 metres per second**.

This is too weak for a small wind turbine. The sun in Mabote is much stronger.

So solar with batteries is cheaper and more reliable here. In a windy place, the tool could add wind.

Short version: "In Mabote the wind is too weak. The sun is strong."

7. How do you measure the risk of power cuts?

In simple ways: **how many hours** the village has no light, and **how many days** have a power cut.

For example: the small system has no light for 653 hours a year. Ours: 27 hours a year.

Short version: "We count the hours and the days without light."

8. What is the difference between efficient and resilient?

Efficient means not wasting money or energy.

Resilient means the village still has light on bad days: clouds, heavy rain, very hot days.

Our system is the **cheapest system that is resilient**. So it is both.

Short version: “Efficient: no waste. Resilient: light on bad days. Ours is both.”

9. Why do you put resilience first, and not efficiency?

Because a cheap system that fails on rainy days is not really cheap.

When the light goes off, the health centre stops, the water pump stops, the children cannot study.

So first we make sure the light stays on. Then we look for the lowest cost.

Short version: “A system that fails on bad days is not a good deal.”

10. Why do you show four days in a row?

Because the battery carries energy from one day to the next.

Sunny days fill the battery. Rainy days use the energy that was stored.

The four days show this clearly: the small system has no light for 17 hours on day 4. Ours has no cuts.

Short version: “Sunny days save energy for rainy days.”

3. Questions about results and money

11. How resilient is your system?

On **98.6 % of the real days**, the village had light all day, with no cuts.

The small system: only 70.6 % of days.

Days with power cuts: from 107 a year to only 5 a year.

Short version: “Light all day on 98.6 % of days.”

12. How much does it cost? Isn't it too expensive?

Our system costs about **81 thousand dollars**. The small system costs 52 thousand.

It is 56 % more, but it is the **cheapest system that keeps the light on**.

Over 20 years, the energy costs **0.17 dollars per kilowatt-hour**. Diesel fuel alone costs 0.43 dollars per kilowatt-hour.

Short version: “It is the cheapest system that keeps the light on.”

13. Your system has extra energy on sunny days. Isn't that a waste?

To fill the battery on a rainy day, we need enough panels. So on sunny days there is extra energy.

This extra energy is useful: the village can use it during the day for the grain mill, water, cooling or phones.

Short version: “The extra energy is free energy for the village to grow.”

14. How long do the batteries last?

We use lithium batteries (LFP). They last about **10 to 15 years**.

We never empty them below 20 %, to protect them. The cost already includes one battery change.

Short version: “10 to 15 years. The cost already includes a new battery.”

4. Questions about impact and the future

15. Can you use this in other villages? (replicability)

Yes. The tool needs only two things: the **sun data** of the place and the **number of families**.

The sun data come from satellites. They are free for any place in the world.

No weather station is needed. The tool is free and open.

Short version: “Yes. Any village: sun data plus number of families.”

16. Can it grow to many villages? (scalability)

Yes. One village takes only a few seconds.

For a whole district, we only need a list of villages and their number of families.

Short version: “Yes. One village takes seconds.”

17. What is the impact on the village?

The health centre, the water pump and the school keep working on rainy days.

No diesel: it avoids about **13 thousand litres of diesel** and **36 tonnes of CO_2** every year.

Short version: “Light, water and health care every day, with clean energy.”

18. Is it possible to build it? (feasibility)

Yes. It uses normal equipment that is sold in Mozambique: about 100 solar panels and lithium batteries.

The tool already works and it is online.

Short version: “Yes. Normal equipment, and the tool already works.”

19. Why did you choose Mabote?

Mabote is a rural district in Inhambane. Most villages have no electricity grid.

It has good sun, but also cloudy and rainy days. It is like many villages in the south of Mozambique.

Short version: “No grid, good sun, but bad days too.”

20. Where do your data come from?

From **ERA5**, a free data set from the European Copernicus programme.

It has the real sun, temperature and wind of Mabote, hour by hour, for thousands of days.

Short version: “Real satellite data from Copernicus, hour by hour.”

21. Is your model tested in the real world?

The model uses standard solar engineering formulas and real weather data.

The village demand is an estimate. The next step is to **measure it in a real village**.

Short version: “The next step is a real test in a village.”

22. What is the next step?

A pilot in one village in Mabote.

We will measure the real energy use for some months and confirm the system size before building it.

Short version: “A pilot in one Mabote village.”

23. What about a cyclone?

We tested the system on real heavy-rain days. That is what happens to the energy during a storm.

Protecting the panels from strong wind is a separate engineering step.

Short version: “We tested real heavy-rain days.”

24. Can your team continue this project?

We are three students from Eduardo Mondlane University. We built the model, the data and the platform ourselves.

For the pilot, we are looking for support from EMERGE and local partners.

Short version: “We built it ourselves, and we want to test it in a village.”

If you don't know the answer, say: “That's a good question. Our model does not cover that yet. It is one of the things we want to check in the pilot.” **Never invent numbers.**

5. How to say the units

Written	Say
kWh	kilowatt-hour(s)
kWp	kilowatt-peak
kW	kilowatt(s)
m/s	metres per second
USD/kWh	dollars per kilowatt-hour
98.6 %	ninety-eight point six percent
81k USD	eighty-one thousand dollars
CO_2	C-O-two (carbon dioxide)